

Mathematics of the Hybrid Triage Engine

Formulas explained simply — with diagrams and flowcharts

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May 19, 2026

Outline

- 1 Big picture
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- 3 Data requirements
- 4 Inter-module interfaces
- 5 Voice and translation pipeline
- 6 Layer 1: NLP and BioBERT
- 7 Layer 2: Deterministic TEWS

What are we building?

In plain English

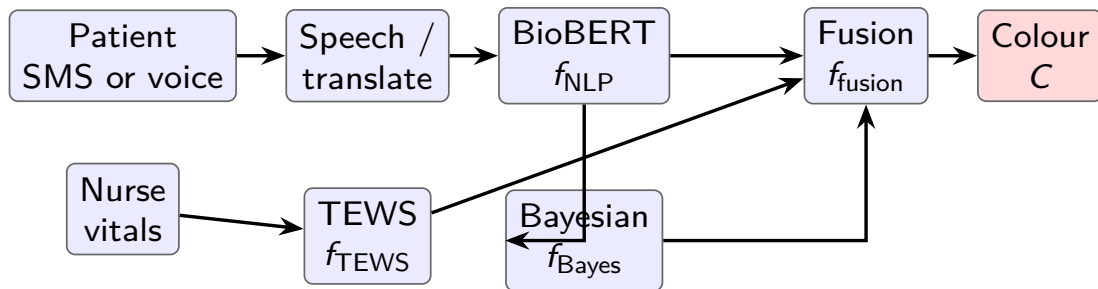
Before any formula: a patient describes symptoms (text or voice). The system must output a **triage colour** (Red / Orange / Yellow / Green) so the hospital knows how urgent the case is.

Three mathematical helpers work together:

- 1 **BioBERT (NLP)** — reads words and finds danger signals in language.
- 2 **TEWS (deterministic)** — adds up points from vital signs (pulse, breathing, etc.).
- 3 **Bayesian (probabilistic)** — guesses urgency when data are missing or confusing.

A **fusion** step combines all three so no single mistake sends a sick patient to a long wait.

Big picture flowchart



Arrows show **data flow**. Fusion runs again whenever text or vitals change.

Why three layers instead of one?

If we only use words (NLP):

- Misses objective physiology
- Can misread slang / translation

If we only use vitals (TEWS):

- Fails when patient has no cuff at home
- Misses “crushing chest pain” with normal pulse

Why we do this

Each layer covers a gap in the others. Mathematics makes the combination **repeatable** and **auditable** for a hospital thesis and for clinical governance.

Symbols — plain language first

Symbol	Meaning in simple terms
X	What the patient said (in English for BioBERT)
D	How strongly each SATS discriminator appears (0 to 1)
T	TEWS danger score from vitals (a whole number)
E	All evidence: symptoms + any measured vitals
P	Probability of each colour given E
C	Final colour the patient is sent to

Next slides use these symbols in formulas.

Global notation

Symbol	Meaning
s	Patient session identifier
t	Event timestamp
X	English-normalised symptom text
X_{tw}	Optional raw Twi transcript
$\mathbf{D} = (d_1, \dots, d_m)$	Discriminator scores, $d_j \in [0, 1]$
$\mathbf{v} = (v_1, \dots, v_n)$	Vital-sign vector (SATS parameters)
T	TEWS score, $T \in \mathbb{Z}_{\geq 0}$
E	Evidence vector (NLP + partial vitals)
$\mathbf{P}$	Posteriors $P(C_k \mid E)$ over colours k
C	Final pathway colour
τ, τ_B	Discriminator / disease override thresholds

Hybrid session state

The engine maintains a session state for each patient s :

$$\mathbf{S}_s = (X, \mathbf{D}, T, \mathbf{P}, C, \text{flags})$$

$$\mathbf{P} = (P(C_{\text{Red}} \mid E), P(C_{\text{Orange}} \mid E), P(C_{\text{Yellow}} \mid E), P(C_{\text{Green}} \mid E))$$

Flags: incomplete_vitals, override_nlp, override_bayes, language code.

Why we do this

We store everything in $\mathbf{S}_s$ so each module reads the same patient story—like one hospital file that updates as new messages or vitals arrive.

Three layers as functions

$$\mathbf{D} = f_{\text{NLP}}(X; \theta_{\text{BERT}}), \quad T = f_{\text{TEWS}}(\mathbf{v}), \quad \mathbf{P} = f_{\text{Bayes}}(E)$$

$$C = f_{\text{fusion}}(\mathbf{D}, T, \mathbf{P}, \text{flags})$$

f_{NLP} : BioBERT encoder + classification head.

f_{TEWS} : piecewise SATS scoring (deterministic).

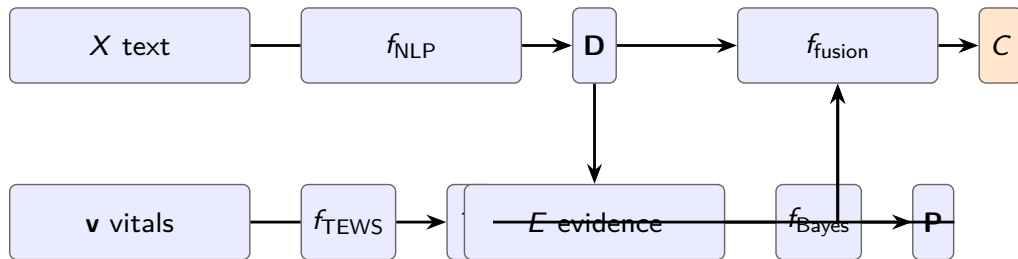
f_{Bayes} : discrete Bayesian network / table lookup.

f_{fusion} : priority rules (Algorithm 1, later).

In plain English

Think of each f . as a **function machine**: input goes in, a well-defined output comes out. That is what makes the chatbot testable in a thesis.

Layer functions flowchart



Evidence vector construction

$$E = \text{concat}(\text{NER}(X), \mathbf{v}_{\text{obs}})$$

- $\text{NER}(X)$: symptom entities and severity tokens from Layer 1
- $\mathbf{v}_{\text{obs}}$: subset of vitals with valid measurements
- Missing v_k omitted from E ; flagged for Bayesian imputation

Why we do this

E is the **shared language** between NLP and Bayes. Symptoms from text and numbers from vitals are combined so probability math sees the whole picture, not just one modality.

Execution order and updates

Parallel (when inputs available):

- $f_{\text{NLP}}(X)$ on every text/voice-derived message
- $f_{\text{Bayes}}(E)$ whenever E changes

Sequential trigger:

- $f_{\text{TEWS}}(\mathbf{v})$ when nurse enters or updates vitals
- f_{fusion} after *any* update to $\mathbf{D}$, T , or $\mathbf{P}$

Re-triage: $\mathbf{S}_s$ is mutable; C may upgrade (e.g. Green $\rightarrow$ Orange) without new patient text.

Urgency ordinal scale

Define urgency ordinals for fusion safety:

$$\text{ord}(\text{Red}) = 4, \text{ord}(\text{Orange}) = 3, \text{ord}(\text{Yellow}) = 2, \text{ord}(\text{Green}) = 1$$

Safety invariant (configured):

$$\text{ord}(C) \geq \max\{\text{ord}(C_{\text{disc}}), \text{ord}(C_{\text{TEWS}}), \text{ord}(C_{\text{Bayes}})\}$$

where C_{disc} is the colour implied by active discriminators in **D**.

Patient and session input data

Field	Type / example
session_id	UUID
language	en, tw
symptom_text or audio	string or blob
age, sex	optional covariates for priors
Self-reported HR, RR	optional v_k (low trust)

Consumed by: ASR, translation, f_{NLP} , partial f_{TEWS} .

Clinical and configuration data

Category	Examples	Used by
Nurse vitals	mobility, RR, HR, temp, AVPU, trauma	f_{TEWS}
SATS tables	$f_k(\cdot)$, w_k	f_{TEWS}
Discriminator list	$\mathcal{D}_{\text{SATS}}$ labels	NLP head
Bayesian priors	$P(C_k)$, $P(E \mid C_k)$	f_{Bayes}
Model weights	BioBERT + classifier θ	f_{NLP}

Audit and governance data

Every session must log (immutable append):

- X_{tw} , X (post-translation), ASR confidence
- Token-level attention highlights (optional interpretability)
- $\mathbf{D}$, T , $\mathbf{P}$, C at each fusion step
- Timestamp and module version IDs

Purpose: Clinical review, bias audits, thesis evaluation metrics.

Why we do this

Without logs you cannot explain **why** the chatbot chose Orange. For a thesis and for KATH governance, every formula output must be reproducible from stored inputs.

Training vs runtime data

Phase	Data
BioBERT pre-training	PubMed + PMC (external; downloaded weights)
Fine-tuning (optional)	KATH-labelled (X , discriminator) pairs
Bayesian calibration	Historical triage outcomes at hospital
Runtime	Single-session inputs only; no batch retrain

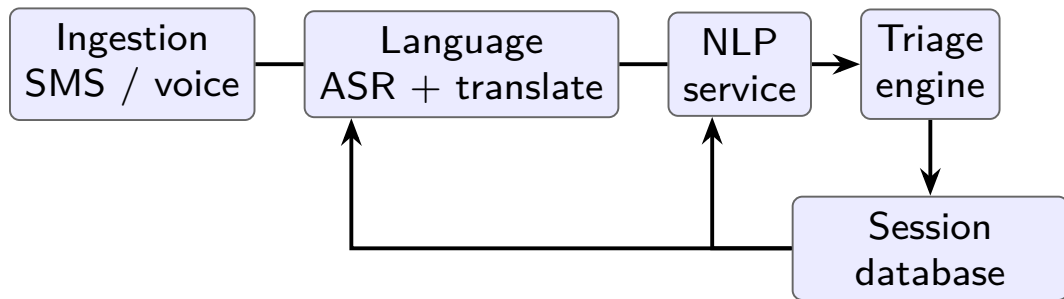
Minimum viable prototype (FYP)

Required: English X + manual vital entry + static SATS tables + fusion code.

Optional modules: voice ASR, Twi translation, live Bayesian DB.

Cross-reference: System narrative deck —
[biobert/integration/hybrid-triage-system.pdf](#).

How software modules talk to each other



In plain English

Modules do not call each other in a tangle—they all read/write one **session record** in a database. That is how we keep voice, text, and nurse vitals in sync.

Session record schema

```
session_id, language, created_at
transcript_tw, transcript_en, asr_confidence
entities: [{label, span, score}]
discriminators: {d_j: float}
vitals: {v_k: float | null}
TEWS: T, tews_incomplete: bool
posteriors: {C_k:  $P(C_k|E)$ }
final_colour: C, fusion_log: [...]
```

Service API sequence

```
POST /speech/recognize -> transcript, language_code  
POST /translate        -> english_text  
POST /nlp/analyze      -> entities, discriminators D  
POST /trriage/compute   -> TEWS T, posteriors P, colour C
```

Each call reads/writes fields in the session record; /trriage/compute runs f_{fusion} .

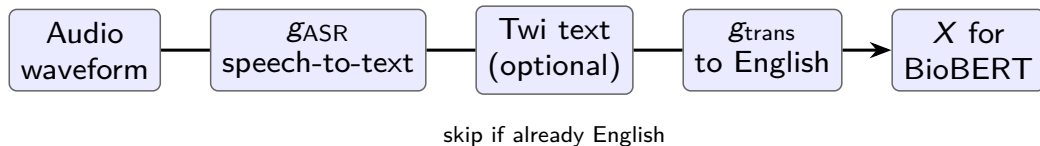
Invocation timeline

Phase 1 (pre-hospital): Patient submits text/voice $\Rightarrow f_{\text{NLP}}, f_{\text{Bayes}}$ on partial $E \Rightarrow$ provisional C .

Phase 2 (triage desk): Nurse enters $\mathbf{v} \Rightarrow f_{\text{TEWS}} \Rightarrow f_{\text{fusion}} \Rightarrow$ final C .

Phase 3 (optional): Re-check vitals $\Rightarrow$ update T , re-run fusion.

Voice pipeline flowchart



Why we do this

BioBERT was trained on biomedical **English**. Twi must become English first, or we use a glossary fallback G when Google Translation does not support Twi.

Formal voice-to-text pipeline

Let $\text{audio} \in \mathbb{R}^N$ be the sampled waveform (numbers describing sound pressure over time).

English path:

$$X = g_{\text{ASR}}(\text{audio}), \quad \text{lang} = \text{en}$$

Twí path:

$$X = g_{\text{trans}}(g_{\text{ASR}}(\text{audio}))$$

Alternatively: $X = G(\tilde{X})$ for glossary fallback G if API unavailable.

In plain English

g_{ASR} turns sound into letters; g_{trans} turns Twí letters into English letters BioBERT can understand.

Google Cloud API parameters

API	Maps to	Billing
Speech-to-Text	g_{ASR}	GCP account; limited free tier
Translation	g_{trans}	$\sim 500\text{k}$ chars/month free (NMT)

Two caveat: ISO code `tw` may not be supported on Translation API — verify at deploy time or use fallback `G`.

Translation error and NLP sensitivity

Model input:

$$X = \hat{X} + \epsilon, \quad \epsilon \text{ translation/ASR noise}$$

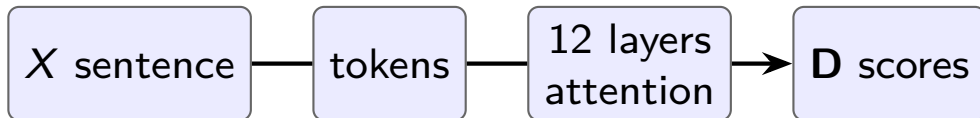
Mitigations:

- Threshold τ on d_j before discriminator override
- Bayesian $P(C_k | E)$ smooths sparse errors
- Human audit of (X_{tw}, X) pairs in pilot

What BioBERT does in one sentence

In plain English

BioBERT reads the patient's sentence and outputs **scores** for clinical danger patterns (SATS discriminators), using the same Transformer math as BERT but trained on medical literature.



Tokenisation

Given $X = w_1 w_2 \cdots w_L$, WordPiece tokeniser τ produces

$$\tau(X) = (t_1, \dots, t_M), \quad M \geq L$$

Special tokens: [CLS] at t_1 , [SEP] for sentence pairs.

Output dimension: $M \leq 512$ (BERT max sequence length).

Why we do this

Computers need discrete tokens, not raw strings. WordPiece splits rare medical words (e.g. "tachycardia") into pieces the model has seen during training.

Input embedding

For each token t_i :

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i)$$

Stack into matrix $H^{(0)} \in \mathbb{R}^{M \times d}$ with $d = 768$ (BERT-Base).

Contextual: E_{word} is a lookup; final $H^{(L)}$ is contextualised by attention.

Why we do this

We add **position** so the model knows word order, and **segment** so it can tell sentence A from sentence B. Without position, “dog bites man” and “man bites dog” would look the same.

Mean pooling vs attention pooling

Mean pooling:

$$\bar{h} = \frac{1}{M} \sum_{i=1}^M h_i^{(L)}$$

Additive attention (DAM-style):

$$a(i) = \frac{\exp(s(h_i; \theta))}{\sum_{j=1}^M \exp(s(h_j; \theta))}, \quad h_{\text{att}} = \sum_{i=1}^M a(i) h_i$$

Clinical example: high $a(i)$ on *chest*, *crushing*; low on *I*, *the*.

Why we do this

Averaging all words treats “I” the same as “crushing”. Attention learns **weights** $a(i)$ that sum to 1, so the model focuses on clinically important tokens—exactly what triage nurses do when listening.

Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

$$Q = HW^Q, K = HW^K, V = HW^V$$

Row i of $\text{softmax}(QK^\top/\sqrt{d_k})$ is the attention distribution of token i over all tokens.

Why we do this

Q, K, V are learned linear transforms of the same sentence. $QK^\top$ measures **how related** two words are; softmax turns that into percentages. Dividing by $\sqrt{d_k}$ stops the numbers from exploding (standard Transformer trick).

Multi-head attention

For head $h \in \{1, \dots, H\}$:

$$\text{head}_h = \text{Attention}(Q_h, K_h, V_h)$$

$$\text{MultiHead}(H) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W^O$$

BERT-Base: $H = 12$, $d_k = d/H = 64$ per head.

Why we do this

Multiple heads let the model notice different things at once—grammar in one head, symptom relations in another—then concatenate and mix with W^O .

Encoder layer /

$$Z^{(l)} = \text{LN}\left(H^{(l-1)} + \text{MultiHead}(H^{(l-1)})\right)$$

$$H^{(l)} = \text{LN}\left(Z^{(l)} + \text{FFN}(Z^{(l)})\right)$$

FFN (per position):

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2$$

BERT-Base: $L = 12$ identical layers.

Why we do this

Residual connections ($+H^{l-1}$) let gradients flow through 12 layers during training. LayerNorm keeps values stable. FFN adds non-linear “thinking” per word after attention mixes information.

MLM pre-training objective

Masked set $\mathcal{M}$ ($\sim 15\%$ of tokens):

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log P(t_i \mid H^{(L)}; \theta)$$

BioBERT: continue MLM on PubMed/PMC corpora starting from BERT weights $\theta_0 \rightarrow \theta_{\text{Bio}}$.

Why we do this

MLM teaches the model to predict hidden words from context—like fill-in-the-blank on millions of medical sentences. That is how it learns **meaning**, not memorisation of one patient message.

Discriminator classification head

Let $h_{[\text{CLS}]} = H_{1,:}^{(L)}$. For discriminator j :

$$d_j = \sigma(w_j^\top h_{[\text{CLS}]} + b_j) \quad \text{or} \quad \hat{\mathbf{y}} = \text{softmax}(Wh_{[\text{CLS}]} + \mathbf{b})$$

Mapping: $\hat{\mathbf{y}} \rightarrow \mathcal{D}_{\text{SATS}}$ (central chest pain, altered consciousness, ...).

Output: $\mathbf{D} = (d_1, \dots, d_m)$.

Why we do this

The [CLS] token is a summary of the whole sentence. We map it to SATS discriminators with sigmoid/softmax so each d_j is a **confidence**, not a free-text diagnosis (safer for patients).

NLP forward chain (inference)

$$X \xrightarrow{\tau} (t_i) \xrightarrow{E} H^{(0)} \xrightarrow{L \times} H^{(L)} \xrightarrow{\text{head}} \mathbf{D}$$

No masking at inference; single forward pass $O(L \cdot M^2 \cdot d)$ per message.

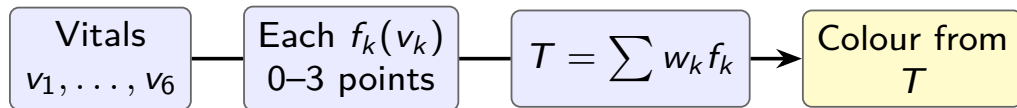
Optional dimensionality reduction

If classifier input dimension d is large:

$$z = W_{\text{PCA}} h_{[\text{CLS}]}, \quad W_{\text{PCA}} \in \mathbb{R}^{r \times d}, \quad r \ll d$$

Then $\mathbf{D} = \text{softmax}(W'z + b')$. Used only if prototype latency requires it.

TEWS flowchart



In plain English

No guessing: same vitals always give the same T . That is why hospitals trust TEWS for audits.

TEWS summation form

SATS Triage Early Warning Score:

$$T = f_{\text{TEWS}}(\mathbf{v}) = \sum_{k=1}^n w_k f_k(v_k)$$

Standard SATS: $w_k = 1$, $n = 6$ parameters:

1. Mobility
2. Respiratory rate
3. Heart rate
4. Temperature
5. AVPU
6. Trauma

Why we do this

We **sum** points because SATS defines triage as additive physiology. Each vital is scored independently then added—transparent for nurses and examiners.

Piecewise score: heart rate f_1

Example mapping for $v_1 = \text{HR (bpm)}$:

$$f_1(v_1) = \begin{cases} 3 & v_1 \geq 130 \\ 2 & 111 \leq v_1 \leq 129 \\ 0 & 51 \leq v_1 \leq 100 \\ 2 & v_1 \leq 40 \\ 1 & \text{otherwise (borderline)} \end{cases}$$

Each f_k is a **step function** — fully deterministic and auditable.

Why we do this

Step functions encode clinical thresholds (e.g. $\text{HR} \geq 130$ is dangerous). There is no neural-network randomness—if a nurse disputes a score, you can point to the table.

Piecewise score: respiratory rate f_2

For $v_2 = \text{RR}$ (breaths/min), analogous thresholds per SATS manual (e.g. $\text{RR} \geq 30 \Rightarrow 3$ points).

Numeric example: $v_1 = 125 \Rightarrow f_1 = 2$; $v_2 = 26 \Rightarrow f_2 = 2$.

Colour map $C_{\text{TEWS}}(T)$

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T > 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases}$$

Example: $f_1 = 2, f_2 = 2, \text{others} = 0 \Rightarrow T = 4 \Rightarrow \text{Yellow}.$

Incomplete vitals

Let $\mathcal{K}_{\text{obs}} = \{k : v_k \text{ measured}\}$.

Partial sum:

$$T_{\text{partial}} = \sum_{k \in \mathcal{K}_{\text{obs}}} w_k f_k(v_k)$$

Set `tews_incomplete` = `true` if $\mathcal{K}_{\text{obs}} \neq \{1, \dots, n\}$.

Fusion must not assign Green on T_{partial} alone when flags indicate missing critical vitals.

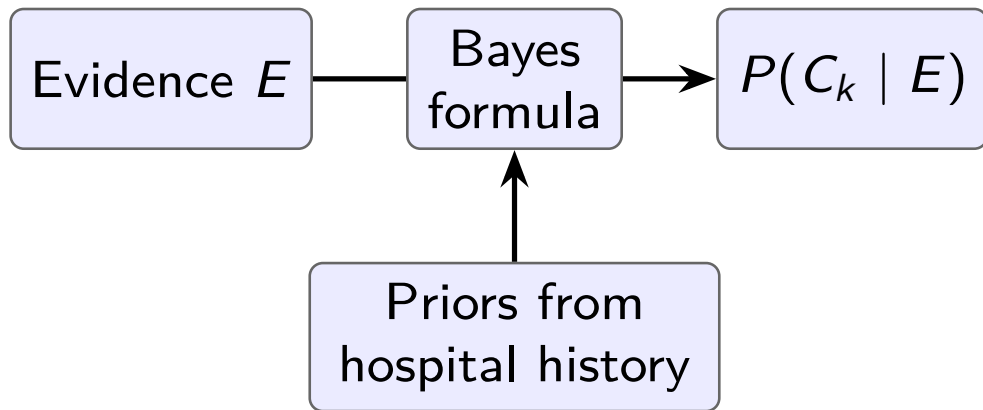
TEWS data contract

Input: vitals map $\{k : v_k\}$ validated against clinical ranges.

Output: $\text{TEWS} = T$, tews_incomplete , $\text{colour_tews} = C_{\text{TEWS}}(T)$.

Consumer: f_{fusion} ; also feeds E for f_{Bayes} .

Bayesian layer flowchart



In plain English

When temperature or BP is missing, we still ask: "Given what we *do* know, which colour is most likely?"

Colour posterior

$$P(C_k | E) = \frac{P(E | C_k) P(C_k)}{\sum_j P(E | C_j) P(C_j)}$$

$C_k \in \{\text{Red, Orange, Yellow, Green}\}$.

$P(C_k)$: hospital prior acuity rates; $P(E | C_k)$: likelihood tables from historical data or expert elicitation.

Why we do this

This is Bayes' rule: **prior** (how common each urgency is) updated by **likelihood** (how often we see these symptoms for that urgency). The denominator normalises so probabilities sum to 1.

Disease-level posterior (override)

For latent severe disease D (e.g. septic shock, MI) and symptom set S from NLP:

$$P(D | S) = \frac{P(S | D) P(D)}{P(S)}$$

Override: if $P(D | S) > \tau_B$ then set $C_{\text{Bayes}} = \text{Red}$ (or Orange per protocol).

Evidence from NLP and vitals

$$E = [\text{entity}_1, \dots, \text{entity}_r, v_{k_1}, \dots, v_{k_s}]$$

Bayesian network structure (conceptual):

- Nodes: C_k , key symptoms, optional vital nodes
- Missing vital node $\Rightarrow$ marginalise or use prior $P(v_k \mid C_k)$

Incomplete evidence branch

When `tews_incomplete` or sparse E :

$$C_{\text{Bayes}} = \arg \max_k P(C_k \mid E)$$

This is the branch used in Algorithm 1 step 5 instead of default Green.

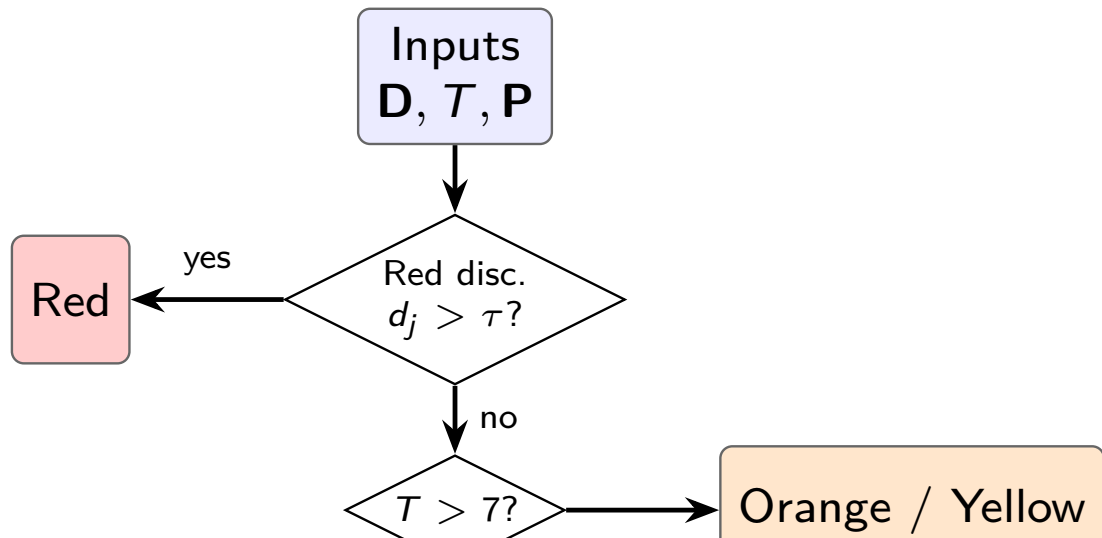
Bayesian data contract

Input: E , prior tables $P(C_k)$, $P(E \mid C_k)$ or $P(S \mid D)$, $P(D)$.

Output: $\mathbf{P}$, `override_bayes`, optional $P(D \mid S)$.

Static data file: `priors_kath.json` (calibrated offline).

Fusion decision flowchart



Algorithm 1: Colour assignment

Input: $\mathbf{D}$, T , $\mathbf{P}$, flags, thresholds τ , τ_B .

- ① If $\exists j : d_j > \tau \wedge j \in \mathcal{D}_{\text{Red}} \Rightarrow C = \text{Red}$.
- ② Else if $\exists j : d_j > \tau \wedge j \in \mathcal{D}_{\text{Orange}} \Rightarrow C = \text{Orange}$.
- ③ Else if $T > 7 \Rightarrow C = \text{Red}$.
- ④ Else if $5 \leq T \leq 6 \Rightarrow C = \text{Orange}$.
- ⑤ Else if $3 \leq T \leq 4 \Rightarrow C = \text{Yellow}$.
- ⑥ Else if incomplete $\Rightarrow C = \arg \max_k P(C_k \mid E)$.
- ⑦ Else if $P(D \mid S) > \tau_B \Rightarrow C = \text{Red}$ (or protocol colour).
- ⑧ Else $C = \text{Green}$.

Conflict resolution

D	T	Result
High (chest pain)	4 (Yellow)	Orange (discriminator)
Low	≥ 7	Red (TEWS)
Ambiguous	incomplete	$\arg \max P(C_k E)$
Low	low	Green

Discriminator and TEWS disagree $\Rightarrow$ take **more urgent** per $\text{ord}(\cdot)$.

Fusion output contract

Output to patient: C , wait guidance, safety message.

Output to hospital: C , $\mathbf{D}$, T , $\mathbf{P}$, full `fusion_log` with rule ID fired (e.g. `RULE_DISC_ORANGE`).

$$f_{\text{fusion}} : (\mathbf{D}, T, \mathbf{P}, \text{flags}) \mapsto (C, \text{log})$$

Module dependency graph (summary)

$$g_{\text{ASR}}, g_{\text{trans}} \rightarrow X \rightarrow f_{\text{NLP}} \rightarrow \mathbf{D}$$

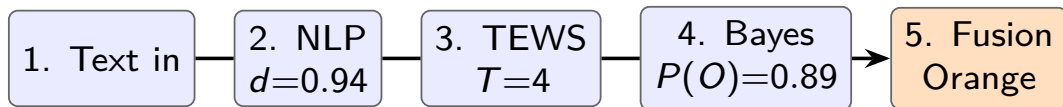
$$\mathbf{v} \rightarrow f_{\text{TEWS}} \rightarrow T$$

$$(\mathbf{D}, \mathbf{v}) \rightarrow E \rightarrow f_{\text{Bayes}} \rightarrow \mathbf{P}$$

$$(\mathbf{D}, T, \mathbf{P}) \rightarrow f_{\text{fusion}} \rightarrow C$$

All modules communicate only through the **session record** schema.

End-to-end trace flowchart



Follow the numbers on the next slides for the chest-pain example.

Trace setup (English SMS)

Input X: “Chest is crushing. Dizzy. Pulse about 125. Breathing 26 per minute. No thermometer.”

Session: lang=en; vitals from text + nurse pending temp/BP.

Goal: compute **D**, **T**, **P**, **C** step by step.

Trace: Layer 1 (NLP)

Token attention α_i : peaks on *crushing*, *chest*, *dizzy*.

$$d_{\text{central_chest_pain}} = 0.94, \quad d_{\text{other}} \ll \tau$$

Entities: chest pain (severe), dizziness, tachycardia (self-report), tachypnoea.
 $\Rightarrow$ **D** triggers discriminator path; *E* enriched for Bayes.

Trace: Layer 2 (TEWS)

From message + nurse: $v_1 = \text{HR} = 125$, $v_2 = \text{RR} = 26$.

$$f_1(125) = 2, \quad f_2(26) = 2, \quad f_{3:6} = 0 \Rightarrow T = 4$$

$$C_{\text{TEWS}}(4) = \text{Yellow}$$

`tews_incomplete=true` (temperature, BP missing).

Trace: Layer 3 (Bayesian)

$E = \{\text{chest pain, dizzy, HR}=125, \text{RR}=26\}.$

$$P(C_{\text{Orange}} \mid E) \approx 0.89, \quad P(C_{\text{Yellow}} \mid E) \approx 0.08$$

$P(\text{MI} \mid S)$ elevated due to symptom cluster.

Trace: Fusion result

Algorithm 1: step 2 fires ($d_{\text{chest}} > \tau$, Orange discriminator).

$$C = \mathbf{Orange}, \quad \text{ord}(C) = 3 > \text{ord}(C_{\text{TEWS}}(4)) = 2$$

Hospital action: cardiology alert; bypass Green/Yellow queue despite $T = 4$.

In plain English

TEWS alone said Yellow ($T = 4$), but the **discriminator** from language said chest pain is an Orange-level SATS finding. Fusion picks the safer, more urgent path.

Trace: Twi voice path

Audio $\rightarrow g_{\text{ASR}} \rightarrow X_{\text{tw}} \rightarrow g_{\text{trans}} \rightarrow X$:

"I have fever, chest hurts, short of breath."

Same chain: $\tau(X) \rightarrow H^{(L)} \rightarrow \mathbf{D}$; fever + chest + dyspnoea $\Rightarrow$ high cardio-respiratory scores.

Fusion identical in structure; $\log(X_{\text{tw}}, X)$ for audit.

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